

Impact of Exchange Rate Pass-Through to South African Prices

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1 Introduction

After over two decades of continuous depreciation, the South African Rand (ZAR) was buoyed by a weaker dollar, higher prices for commodities like gold and platinum-group metals and improved investors sentiment towards the end of 2025. At around the same time, the National Treasury and the South African Reserve Bank (SARB) announced the move from the target range of 3%-6% to a 3% target with a 1% tolerance band. This change in monetary policy is aimed at bringing down expected and actual inflation as well as borrowing costs (**sa'infl**). In order to maintain credibility, the Reserve Bank will need to be vigilant of potential drivers of inflation, both in the domestic as well as the global economy. A rebound in the value of the dollar, cooling global demand for commodities and fickle investor sentiment could all trigger further depreciation of the Rand at the expense of price stability. Indeed, while South African exports have benefitted from a weaker exchange rate, deindustrialization and the associated decline in export volume and complexity means that gains have been limited to the agricultural and commodities sector (**stern'ramkolowan**). In addition, the continued deindustrialisation and disappearance of key upstream inputs may lead to an increasing reliance on imports for inputs and final goods. Given these facts, determining the extent to which domestic prices are influenced by the exchange rate is of concern to policymakers aiming for low inflation.

While the floating exchange rate may act as a shock absorber through responding to real shocks, structural factors like market imperfections and price rigidities may render the absorption incomplete. Such a shock to the exchange rate may improve a country's terms of trade through bolstering exports and dampening imports, however this outcome depends on the elasticity of substitution between domestic and foreign goods. This, in turn, depends on the degree to which a country depends on foreign goods and vice versa - if a country is unable to substitute its imports of foreign goods while at the same time not enjoying foreign demand for its domestic products, an exchange rate depreciation will lead to a deterioration in a country's trade balance and higher domestic prices (**obstfeld1995exchange**). This indirectly relates to the degree of local currency pricing (LCP) and producer currency pricing (PCP), referring to goods priced in destination and origin country currencies respectively. Concretely, if goods are priced in the

domestic currency, nominal rigidity means there would be no impact of exchange rate fluctuations on domestic prices, while conversely goods being priced in the currency of the foreign country means that fluctuations lead to corresponding changes in domestic prices. According to **devereux2004****endogenous**<empty citation>, producers' pricing decisions are made according to the variance of the exchange rate and its covariance with marginal costs: if firms face higher exchange rate volatility, they have a stronger incentive to price according to their own currency, while a higher covariance between marginal costs and exchange rate. As a result, currencies with lower variability are more likely to be chosen as the pricing medium.

While exchange rate stability may explain differences in pass-through across countries, industry-level features are also likely to determine currency pricing and thus pass-through. If an exporting firm has little market share in a given industry in the foreign market, their optimal strategy would be to follow LCP in order to avoid swings in demand owing to higher degrees of elasticity (**NBERw9039**). To the extent that there is variation in trade patterns across different industries, there is likely to be heterogeneity in pass-through among different parts of the economy.

Exchange rate pass-through has previously been quantified in a variety of papers. **campa2005****exchange**<empty citation> investigate the elasticity of import prices with respect to exchange rates across 23 countries using ordinary least squares, finding estimates between 0.19 and 0.79 in the short run (1 quarter after) and higher values in the long-run (4 quarters). In addition to the cross-country analysis, the authors observe the evolution of pass-through across time, finding changes in pass-through for their sample of countries to be tied closer to changes in the composition of import bundles than changes in macroeconomic variables, although in the cross-country analysis, they find inflation to explain level differences in pass-through. In **choudhri2015****exchange**<empty citation>, the authors develop a DSGE model with Calvo pricing and a proportion of firms using PCP ranging between 0 and 1 to estimate the effect of an exchange rate shock on import prices. The authors find this model to be coherent with real-world data estimated using a vector autoregression, with pass-through estimates varying positively with the degree of PCP. Despite being able to estimate the effect of exogenous shocks on the exchange rate, this approach is not able to identify the sources of shocks and how prices respond to different shocks. An earlier study by **shambaugh2008****new**<empty citation> makes use of long-run restrictions to identify the effect of different types of shocks on the exchange rate and prices respectively before computing pass-through as the ratio of the change in price to the change in the exchange rate.

While past studies have investigated pass-through differences across countries and industries internationally, there is limited evidence on pass-through differences within specific countries. This is surprising given the policy relevance and the aforementioned literature evincing the likelihood of different pass-through rates based on industry characteristics. Through exploring the response of various domestic prices to shocks in the exchange rate, this paper will elucidate the relationship between the exchange rate and different parts of the economy, allowing for a more nuanced understanding of pass-through.

2 Literature Review

Since the 1980's, literature on the show that the relationship between exchange rates and prices can be assessed through both micro and macroeconomic perspectives. With regard to the former, industrial organization theory has been used to explain disconnects between changes to nominal exchange rates and prices and the resulting violation of the Law of One Price over the long term. According to **dornbusch1985exchange**<empty citation>, exchange rate movements interact with sticky prices and wages, producing a cost-shock for foreign firms in domestic markets which will result in an industry-wide adjustment in prices based on the industry's market concentration, producer homogeneity, substitutability and relative split between foreign and domestic goods. **krugman1986**<empty citation> provides three potential explanations for why dynamic exchange rates may not lead to proportionate adjustments in import prices. Supply-side constraints (for example in marketing and distribution channels) in different markets dampens the effect of a depreciation in the domestic currency on domestic prices, as foreign firms face costs in expanding supply infrastructure. On the demand-side, to the extent that the impact of prices on demand is lagged, foreign producers anticipating domestic currency depreciations may decrease their current prices in order to avoid losing sales. Finally, firm reputation concerns may also affect pricing decisions when agents face additional costs to reputation among importers when changing prices. **marston1990pricing**<empty citation> studied the behaviour of Japanese firms exporting to the US and showed that differences in the elasticity of mark-ups between domestic and foreign markets and interdependence in quantities allocated to foreign and domestic markets mean that yen depreciations can lead to exporting firms increasing their mark-ups, resulting in a decrease in the dollar price that is less than the magnitude of the dollar appreciation.

Other studies have modelled exporting firms' currency pricing decisions and pass-through according to macroeconomic features. In the partial equilibrium models by **friberg1998currency**<empty citation> and **engel-2006**<empty citation>, currency volatility and its covariance with costs and demand structures facing firms affect their decisions about which currencies to invoice in, with **devereux2004endogenous**<empty citation> assuming that higher exchange rate volatility incentivises PCP when production costs are local, leading to higher ERPT. **bacchetta2005theory**<empty citation> map out a general equilibrium model showing that pricing decisions are informed by relative market size and product differentiation, with exporters being more likely to invoice in currencies with large and differentiated markets. Alongside exchange rate dynamics, inflation can also play a role in determining how exchange rate changes are passed through. According to **taylor2000low**<empty citation>, that the low-inflation environment of the 1990's coincided with a fall in ERPT can be explained in part by increasing international competition and lower trade barriers reducing firms' pricing power as well as by a virtuous but fragile cycle of low inflation expectations, which means that importing firms do not raise prices as marginal cost increases resulting from, for example an increase in the nominal exchange rate, are not anticipated to be permanent. A similar argument is made in **gagnon2004monetary**<empty citation>, who show that lower inflation and inflation targeting by monetary authorities both contribute to lower levels of pass-through to import prices. **ozkan2015time**<empty citation> drew similar conclusions from studying emerging market economies, noting in addition the negative relationship between ERPT and exchange rate volatility and trade openness.

In the seminal international macroeconomic models by **obstfeld1995exchange**; **obstfeld1998risk**<empty citation>

producers set prices in their own currencies (producer-currency pricing), meaning that the prices of imported goods should change one-for-one in response to exchange rate fluctuations and that these in turn help re-establish market equilibria. To account for the empirical evidence documented in **goldberg1997causes**<empty citation> that import prices in industrialised countries tend not to move with the exchange rate, **devereux2003monetary**<empty citation> introduce an alternative model in which exporting producers set prices in the currency of the destination country (local-currency pricing), meaning that exchange rate fluctuations have no impact on the prices of imported goods. This in turn they show to dampen the expenditure-switching benefits of a floating exchange rate, a point reinforced by **devereux2004endogenous**<empty citation> and **betts2000exchange**<empty citation> allow for a mix between LCP and PCP, showing that under sticky prices, the response of prices to exchange rate movements is a function of the share of domestic firms and foreign firms who price to market (LCP) over all firms.

As posited in **betts2000exchange**<empty citation> and **devereux2003monetary**<empty citation>, currency pricing decisions have implications for the frequency of price changes and degree of pass-through. A study of US import transactions by **gopinath2010currency**<empty citation> showed large differences in pass-through between goods priced in Dollars and non-Dollar currencies, with the latter showing higher responses to exchange rate changes even at long horizons. The authors find a strong positive correlation between the share of PCP imports from a partner and the country's average bilateral pass-through into US import prices, and point to similar dynamics as those in **dornbusch1985exchange**<empty citation> and **krugman1986**<empty citation> to explain persistent differences in pass-through between LCP and PCP imports.

Empirical investigations into ERPT shows import invoicing to be a mix of PCP and LCP. Across 25 OECD countries, **campa2005exchange**<empty citation> use a multivariate OLS controlling for costs and demand factors instead of simply measuring the effect of the change in the exchange rate on import prices. They found the average pass-through of exchange rate to import prices to be about 0.6 after a quarter and 0.75 after a year, although these values differ across countries. **colavecchio2019non**<empty citation> use linear projection models to investigate ERPT to import prices in the Euro area between 1997 and 2019, finding an average pass-through of 0.20 after one quarter and 0.33 after a year, with non-linear effects of magnitude found, but no difference in ERPT between appreciations and depreciations. A handful of studies have examined ERPT to import prices in South Africa, with **aronetal2012**<empty citation> employing the approach used by **campa2005exchange**<empty citation> along with single- and multi-equation error correction models, finding pass-through to import prices to be around 0.43-0.5 after a year, though this estimation was lower in the second part of the full sample. Other studies on South Africa employing single-equation error-correction models found very high pass-through coefficients to import prices between 0.75 and 0.93 **Karoroetal2008**; **kabundi2016exchange**.

mccarthy1999<empty citation> conceptualised a multi-equation model with a Cholesky decomposition to estimate ERPT. Using the residuals of exchange rate movement regressed on supply and demand surprises to remove some endogeneity, the ERPT onto import, PPI and CPI are subsequently measures one after the other. The paper indicates shows decreases in import prices after a 1% exchange rate depreciation, although this is found to decrease and become less precise after a year. A similar response of import prices to exchange rate changes among G-7 countries is reported in **choudhri2005explaining**<empty citation>, which finds strong initial responses of import prices, which decline after a year, creating a hump-shaped

response curve. In a subsequent paper, **choudhri2015exchange**<empty citation> estimate first-quarter pass-through for a variety of countries including South Africa, finding ERPT to be 0.31, with estimates in the second half of the sample (1998- 2010) to be higher than in the preceding twelve years (0.38 vs 0.2). Studying ERPT into the Euro area using this methodology, **hahn2003**<empty citation> found non-oil import prices to be 20

The use of such a methodology is advantageous not only in its incorporation of dynamic relationships between variables, but also because it allows for the analysis of how downstream prices are affected by exchange rate shocks. Local transaction and distribution costs as well as differences in market structures facing firms at each point in the distribution chain can cause deviations in final consumer price indices from exchange rate movements. **burstein2003distribution**<empty citation> showed substantial distribution margins in the US and Argentina, while **goldberg2010sensitivity**<empty citation> found that distribution margins dampened the effect of exchange rates on consumer prices across a sample of advanced economies, with the distribution costs themselves a function of the input of imported goods into the non-tradeable distributive goods or services. **engel2002responsiveness**<empty citation> argues that producers face a sourcing problem between domestic and imported inputs, which influences the ability of importers to pass through price changes to producers of final goods. The disconnect between ERPT to import prices and consumer prices may also be the result of by the contribution of non-tradables to producer and consumer baskets **obstfeld2000new**<empty citation>. To the extent that two countries differ only in their degree of self-sufficiency, the more self-sufficient country will have lower pass-through to consumer prices.

Empirical evidence of dampened pass-through from import prices to consumer prices. **mccarthy1999**<empty citation> found lower pass-through across the nine country sample and showed that the contribution of modelled variables to the variance decreased from import prices to PPI and CPI, demonstrating the increasing primacy of other factors in determining latter prices. In the variance decomposition for prices, **hahn2003**<empty citation> finds oil shocks to be the most important contributing factor to the variance, with the contribution of exchange rate shocks to variance falling from around 15% after a year and a half for import prices, and decreasing to between 5% and 10% over the same timeframe for consumer prices. PPI and CPI curves are found to be more muted in the model specifications by **choudhri2005explaining**<empty citation> and lower ERPT estimates are reported for PPI and CPI in the cross-country study in **choudhri2005explaining**<empty citation>. Across all of these studies, it is evident that responses occur less rapidly and with less magnitude as one moves down the pricing chain, an analogous finding to the impact of monetary shocks reported in **clark1999responses**<empty citation>. For South Africa, **kabundi2016exchange**<empty citation> find the second-stage pass-through (to consumer prices) to be 36% compared to around 90% for first stage pass-through in their single-equation model, while **parsley2012exchange**<empty citation> estimate ERPT to consumer prices at 14% to 27% in a panel specification. Unlike **jooste2014determinants**<empty citation> whose time-varying BVAR specification finds inflation variability to contribute to higher pass-through to consumer prices, **parsley2012exchange**<empty citation> argues that observed declines in ERPT to consumer prices are the result of a changing trade composition.

The extensive literature on ERPT and adjacent topic enables a thorough exploration of the topic at hand. This study of ERPT to South Africa prices contributes to the literature in the following regards. Firstly, applying a multi-equation model to the South African context, in contrast to previous studies which have used simple-equation models, allows for richer dynamic analysis as previously done in **mccarthy1999**<empty citation> and **hahn2003**<empty citation> that also incorporates feedback effects among variables. Secondly, the availability of a longer

horizon since the SARB adopted and succeeded in inflation targeting allows for a renewed, better-informed assessment of monetary regime dependence and the role of inflation in determining pass-through.

3 Methodology

The identification framework employed in this paper is based on the methodology developed by **mccarthy1999**<empty citation> and subsequently used by **choudhri2015exchange**<empty citation>. In this model, a distribution chain of price shocks from import prices to producer prices to consumer prices is used. The prices at each stage are in turn determined by the expected prices based on information available at the end of period $t - 1$, contemporaneous supply, demand and exchange shocks, as well as the inflation shocks at preceding points in the distribution chain. This implies a recursive structure where supply conditions affect demand, both of which influence the exchange rate, which together impact the domestic prices along the distributional chain. For each stage in the recursive sequence, there is no contemporaneous feedback from subsequent variables in the model. In economic terms, this means we assume prices do not immediately affect the exchange rate and is based off the fact that while exchange rates can be observed at any given moment, data on prices are published at discrete frequencies and thus are reported with a lag **choudhri2005explaining**. The primacy of oil shocks as an upstream disturbance is supported by **kilian2009** and **Kanzig**, who argue that oil supply does not respond to demand within the same month, and that supply decisions taken by OPEC countries are to a large extent exogenous. Formally, this model can be represented by the following system of equations:

$$\pi_t^{oil} = E_{t-1}(\pi_t^{oil}) + \varepsilon_t^{oil} \quad (1)$$

$$\tilde{y}_t = E_{t-1}(\tilde{y}_t) + a_1 \varepsilon_t^s + \varepsilon_t^d \quad (2)$$

$$\Delta e_t = E_{t-1}(\Delta e_t) + b_1 \varepsilon_t^s + b_2 \varepsilon_t^d + \varepsilon_t^e \quad (3)$$

$$\pi_t^m = E_{t-1}(\pi_t^m) + \alpha_1 \varepsilon_t^s + \alpha_2 \varepsilon_t^d + \alpha_3 \varepsilon_t^e + \varepsilon_t^m \quad (4)$$

$$\pi_t^w = E_{t-1}(\pi_t^w) + \alpha_1 \beta_t^s + \alpha_2 \beta_t^d + \alpha_3 \beta_t^e + \alpha_4 \beta_t^m + \varepsilon_t^w \quad (5)$$

$$\pi_t^c = E_{t-1}(\pi_t^c) + \alpha_1 \gamma_t^s + \alpha_2 \gamma_t^d + \alpha_3 \gamma_t^e + \alpha_4 \gamma_t^m + \alpha_5 \gamma_t^w + \varepsilon_t^c \quad (6)$$

Where (ε_t^j) represents an orthogonalized shocks in variable j during period t .

The main quantitative estimation of the impact of exchange rate shocks on domestic prices applies a vector autoregressive model introduced by **simms1980**<empty citation> and used extensively in subsequent macroeconomic literature. Combining this with the recursive identification framework set out by **mccarthy1999** means that we impose restrictions on the relationships between variables, allowing for estimation using a structured VAR analogous to that developed in **BlanchardWatson1986**<empty citation>, in which estimate disturbances in variables from previous equations are used as instruments to estimate disturbances for each equation. This set-up requires that disturbances obtained in each equation be uncorrelated with

each other. Although it would be unrealistic to assume that disturbances are completely independent, the correlations between disturbances in the system are not found to be significant enough to warrant concern¹. Following **sims1986forecasting**<empty citation> where we have a structural model that can be denoted as:

$$AY_t = \sum_{l=1}^p C_l Y_{t-l} + \varepsilon_t \quad (7)$$

where Y_t is a vector with $Y_t' = (\pi_t^{oil}, \tilde{y}_t, \Delta e_t, \pi_t^m, \pi_t^w, \pi_t^c)$, C_l is a matrix of coefficients of Y_{t-l} on Y_t , ε_t is a vector of structural shocks at t and A is the matrix of contemporaneous coefficients. The recursive structure modelled by equations (1) - (6) constitutes a Wold causal chain, meaning A^{-1} is a triangular matrix with a diagonal populated by ones representing the variance of each (normalised) variable and off-diagonals representing the contemporaneous effects among different variables. In this matrix, the j -th column represents the immediate impact of a shock to the j -th variable in Y_t

Multiplying both sides of (7) by A^{-1} , yields:

$$Y_t = \sum_{j=1}^p A^{-1} C_l Y_{t-l} + A^{-1} \varepsilon_t \quad (8)$$

.

Given that A^{-1} is a lower-triangle (by the Wold causal chain), structural disturbances in ε_t affect only the variables ordered below them in Y_t . Thus, π_t^{oil} is affected only by historical values of Y_t and the contemporaneous value of ε_t^{oil} while π_t^c is impacted by contemporaneous structural disturbances of all variables.

Using the fact that the variance-covariance matrix Σ is a positive definite along with the fact that errors in ε_t are normalised and orthogonal, a Cholesky decomposition can be applied to recover A^{-1} , the matrix of contemporaneous coefficients of structural disturbances, by:

$$\Sigma = E[A^{-1} \varepsilon_t (A^{-1} \varepsilon_t)'] \quad (9)$$

$$= E[A^{-1} \varepsilon_t \varepsilon_t' (A^{-1})'] \quad (10)$$

$$= E[\varepsilon_t \varepsilon_t'] A^{-1} (A^{-1})' \quad (11)$$

$$= A^{-1} (A^{-1})' \quad (12)$$

$$\text{where } E[\varepsilon_t \varepsilon_t'] = I \quad (13)$$

The object of interest in this paper is the effect of exchange rate shocks on prices. This can be done through recovering the impulse response function:

¹See Appendix

$$\theta_{VAR}^h = \varepsilon_{c/w}' B_h A^{-1} \varepsilon_e, \quad (14)$$

$$\text{where } B_h = \sum_{l=1}^{\min\{p, h\}} (A^{-1} C_l) B_{h-l} \quad (15)$$

where the coefficient θ_{VAR}^h captures the dynamic effect of a exchange rate shock propagated through (8) recursively over h periods. The recursive structure of the identification scheme from (1) - (6) implies knowledge the form of the data generating process is held to be known. Through using a VAR model, dynamic interactions between variables can be more precisely estimated, however to the extent that the VAR dynamics diverge from reality, estimates may suffer from misspecification bias. To address this possibility, the main results from the SVAR are compared with results obtained from local projections as per **Jorda2005**<empty citation>.

This approach involves regressing our prices at each period across horizon h on an initial exchange rate disturbance along with a set of controls reflecting the posited structural relationships between the variables using ordinary least squares. This regression yields less precise estimates that reflect additional uncertainty around the data generating process, however is less likely to suffer from misspecification bias as each period is regressed independently instead of using the recursive SVAR estimation. Using sufficiently informative lags in our SVAR model should, however, reduce this possibility, although using higher lag orders does come at a substantial cost to the degrees of freedom in the SVAR. Although main results will be obtained using the SVAR model, discussion can be supplemented through comparing the estimates with those from LP.

In addition, while standard VAR and LP models are sign-agnostic in that disturbances are assumed to have symmetric effects (for example, a 1% appreciation has the exact opposite impact as a 1% depreciation),

Although Akaike Information Criteria can be used to select the lag order, the tendency of AIC to pick out more parsimonious models with shorter lag lengths risks longer-horizon dynamics being omitted. Oil shocks, for example, can have gradual or delayed impacts on economic variables, meaning that short lag lengths may miss these effects altogether **EDELSTEIN2009766; hamilton-herrera**. Following **mccarthy1999**<empty citation>, lag order is restricted to a year. This depth provides a balance between, on one hand maintaining greater degrees of freedom and on the other capturing long-term dynamics and is a standard application in VAR models **Lutkepohl**. Applying $p = 12$ across both SVAR and LP models implies that estimates will be similar between the two within the first year of the shock, with differences between the two models manifesting beyond this window.

4 Data

For the purposes of this study, monthly data from February 1990 up to December 2022 are used, thus spanning a total of 395 months. Along with looking at the full sample, the sample

is divided into two periods: 1990 up to and including 2008 and 2009 until the end of 2022. Although this results in a slightly unbalanced comparison with the early period consisting of 227 months and the latter of 168, 2009 did mark a turning point for South Africa with lower average and variance in the inflation rates, representing a period of greater price stability. This in turn allows for the assessment of whether ...

Oil shock data is obtained from **Kanzig**², who uses changes in oil futures prices during tight windows around OPEC announcements to construct an oil supply shock series, where the effect of confounding structural shocks on oil supply is purged by controlling for changes in OPEC's global demand forecast, a technique analogous to that used in **romer**³. This series is particularly useful when applied to a small economy like South Africa, as the likelihood of any feedback from domestic disturbances to oil shocks is negligible, making it a superior approach to that used in **mccarthy1999**, who uses residuals of domestic oil price dynamics. Related to the fact that it is a more exogenous series than domestic oil prices, and that as a result the Känzig oil supply shock series has the significant benefit to both VAR and LP models of fixing coefficients on oil shocks to zero. Shocks are entered into the model as fully exogenous.

A limitation imposed by this approach is that the impact of oil (supply) shocks on prices cannot be recovered via forecast error variance decomposition. This is a non-trivial issue given the primacy of oil shocks on downstream economic variables, however despite the centrality of oil shocks to the economy, it does not form part of the central undertaking of this study. Nevertheless, a fully endogenised model is included as part of the sensitivity analysis. As mentioned, however, the series is exogenous by construction, particularly in relations to a small economy, thus endogenising the series risks miscalculating the size of the shock if parts of it are "explained" by spurious correlation with previous values of Y_{t-j} . While this leads to a deviation from previous literature, it decreases the bias that may result from spurious correlation with endogenous variables and increases the degrees of freedom in the estimation.

In order to estimate demand shocks, seasonally-adjusted quarterly Real GDP figures are interpolated to a monthly frequency using Denton-Cholette interpolation, which allows for temporal disaggregation in the absence of any suitable high-frequency indicator variables³ through distributing quarterly values evenly across constituent months by minimizing period-to-period differences subject to the constraint that three-month values should sum to quarterly values. The Denton-Cholette method also avoids the starting point problem present in the original formulation by **denton1971**, which imposes no constraint on the first high-frequency point from the previous period, resulting in a potential discontinuity at the starting point as, unlike interior data points, there is only one low-frequency constraint **DagumCholette2006**. Because this adjustment does not account for seasonal trends, instead distributing quarterly values smoothly, using seasonally-adjusted data means that seasonal variation present in unadjusted series is not lost. A Hodrick-Prescott filter is applied to the disaggregated and logged series, from which the cyclical component, or the deviation from trend, is recovered and used as the output gap.

²Updated vintages are available online at: <https://github.com/dkaenzig/oilsupplynews>

³Regression-based methods like Chow-Lin and indexed Denton-Cholette were not suitable as the Bureau of Economic Research's Business Cycle Index series was not found to be sufficiently correlated with GDP series.

The SARB's Nominal Effective Exchange Rate⁴, which measures the external value of the Rand against a weighted basket of currencies is the preferred measure for the exchange rate⁵. Although currencies are weighted by partners' shares of imports and exports, it is preferred to the Dollar given the importance of other major economies like China, the UK and the EU as trade partners and has precedents in previous studies⁶. Indexed values are log-differenced and flipped such that positive movements represent depreciations.

South African import data are characterised by various breaks which compromise its suitability: between 1980 and 2012, import prices formed part of producer price indices and were subdivided by Standard Industry Classification. After 2010, import prices were obtained from customs data using a classification system based on the Central Product Classification System, which aggregates goods together according to physical characteristics. This index was calculated as the arithmetic mean up until 2023, while a geometric mean was used from 2017 up to the time of writing **StatsSA2025XUVI**. Tests were carried out to determine the compatibility between the different series.

The overlapping period between 2010 and 2013 also indicates that the two series

For producer and consumer prices, headline PPI and CPI figures are used. All price series are log-differenced.

Results of diagnostic tests are provided in the ???. Series are found to be stationary according to Augmented Dickey-Fuller tests, thus justifying the use of the VAR model as opposed to a vector error correction model. Residuals from the VAR models display very low correlation between different variables and across lags. Normality tests indicate that our data displays skewness, with Jarque-Bera p-values being significant beyond 1% for all variables. With this in mind, series are bootstrapped?????

⁴a detailed description of the methodology is available at <https://www.resbank.co.za/content/dam/sarb/publications/quarterly-bulletins/quarterly-bulletin-publications/2020/10091/03Note-on-the-revision-of-South-Africas-nominal-and-real-effective-exchange-rate-indices.pdf>

⁵The USD-ZAR exchange rate is used as a robustness check

⁶See **mccarthy1999**<empty citation>, **Karoroetal2008**<empty citation>, **aronetal2012**<empty citation>, **choudri2015exchange**<empty citation>, **kabundi2016exchange**<empty citation>

5 Results

5.1 Impulse Responses

5.2 Variance Decomposition

Table 1: Forecast Error Variance Decomposition at Selected Horizons

		Horizon (months)				
Sample	Imports	3	6	9	12	15
Import Price Index						
1990 - 2023	Excl. Imports	NA	NA	NA	NA	NA
	Incl. Imports	0.249	0.252	0.241	0.238	0.237
1990 - 2010	Excl. Imports	NA	NA	NA	NA	NA
	Incl. Imports	0.172	0.196	0.186	0.180	0.182
2010 - 2023	Excl. Imports	NA	NA	NA	NA	NA
	Incl. Imports	0.465	0.417	0.391	0.381	0.373
Producer Price Index						
1990 - 2023	Excl. Imports	0.088	0.124	0.122	0.125	0.125
	Incl. Imports	0.087	0.120	0.118	0.120	0.119
1990 - 2010	Excl. Imports	0.079	0.134	0.129	0.132	0.133
	Incl. Imports	0.074	0.131	0.124	0.126	0.127
2010 - 2023	Excl. Imports	0.182	0.169	0.166	0.161	0.158
	Incl. Imports	0.190	0.180	0.175	0.169	0.166
Consumer Price Index						
1990 - 2023	Excl. Imports	0.056	0.064	0.075	0.091	0.095
	Incl. Imports	0.059	0.065	0.075	0.087	0.089
1990 - 2010	Excl. Imports	0.071	0.083	0.090	0.113	0.115
	Incl. Imports	0.081	0.091	0.098	0.114	0.115
2010 - 2023	Excl. Imports	0.039	0.061	0.070	0.073	0.075
	Incl. Imports	0.038	0.062	0.073	0.074	0.073

6 Appendix

6.1 Results

Table 2: Cumulative Exchange Rate Pass-Through at Selected Horizons

Horizon	Excl. Imports			Incl. Imports		
	1990 - 2023	1990 - 2010	2010 - 2023	1990 - 2023	1990 - 2010	2010 - 2023
Import Price Index						
3	0.5 (0.07)	0.43 (0.08)	0.65 (0.12)	NA	NA	NA
6	0.57 (0.1)	0.48 (0.11)	0.6 (0.18)	NA	NA	NA
9	0.52 (0.14)	0.38 (0.13)	0.61 (0.24)	NA	NA	NA
12	0.48 (0.14)	0.34 (0.15)	0.58 (0.28)	NA	NA	NA
15	0.47 (0.15)	0.25 (0.15)	0.62 (0.32)	NA	NA	NA
Producer Price Index						
3	0.15 (0.03)	0.15 (0.04)	0.19 (0.05)	0.15 (0.03)	0.15 (0.04)	0.18 (0.05)
6	0.21 (0.05)	0.23 (0.05)	0.2 (0.07)	0.21 (0.05)	0.24 (0.06)	0.17 (0.06)
9	0.23 (0.06)	0.26 (0.07)	0.23 (0.08)	0.23 (0.06)	0.26 (0.08)	0.2 (0.08)
12	0.25 (0.07)	0.27 (0.08)	0.24 (0.08)	0.25 (0.06)	0.26 (0.09)	0.2 (0.08)
15	0.24 (0.07)	0.23 (0.08)	0.22 (0.08)	0.25 (0.06)	0.23 (0.1)	0.19 (0.09)
Consumer Price Index						
3	0.08 (0.02)	0.12 (0.03)	0.02 (0.02)	0.08 (0.02)	0.11 (0.03)	0.02 (0.02)
6	0.12 (0.03)	0.16 (0.05)	0.05 (0.03)	0.12 (0.03)	0.15 (0.05)	0.06 (0.03)
9	0.15 (0.04)	0.21 (0.07)	0.07 (0.03)	0.16 (0.04)	0.2 (0.06)	0.07 (0.03)
12	0.18 (0.05)	0.26 (0.08)	0.07 (0.03)	0.19 (0.05)	0.26 (0.08)	0.07 (0.03)
15	0.2 (0.06)	0.27 (0.09)	0.04 (0.03)	0.21 (0.06)	0.27 (0.09)	0.05 (0.03)
	329	174	91	339	183	100

Note: Cumulative impulse responses to a one percent exchange rate depreciation.

Bootstrapped standard errors in parentheses (500 replications).

6.2 Diagnostics

6.2.1 Stationarity

Table 3: Descriptive Statistics

Variable	1990 - 2023	1990 - 2010	2010 - 2023
Oil Shock	-0.009 (0.589)	0.001 (0.58)	-0.025 (0.604)
Demand	0 (0.006)	0 (0.003)	0 (0.008)
Exchange Rate	0.004 (0.03)	0.005 (0.031)	0.004 (0.029)
Importer Price Index	0.003 (0.016)	0.005 (0.016)	-0.001 (0.016)
Producer Price Index	0.005 (0.007)	0.006 (0.008)	0.005 (0.006)
Consumer Price Index	0.005 (0.005)	0.006 (0.006)	0.004 (0.004)

Note: Mean (standard deviation). Sample splits at 2010-02-01.

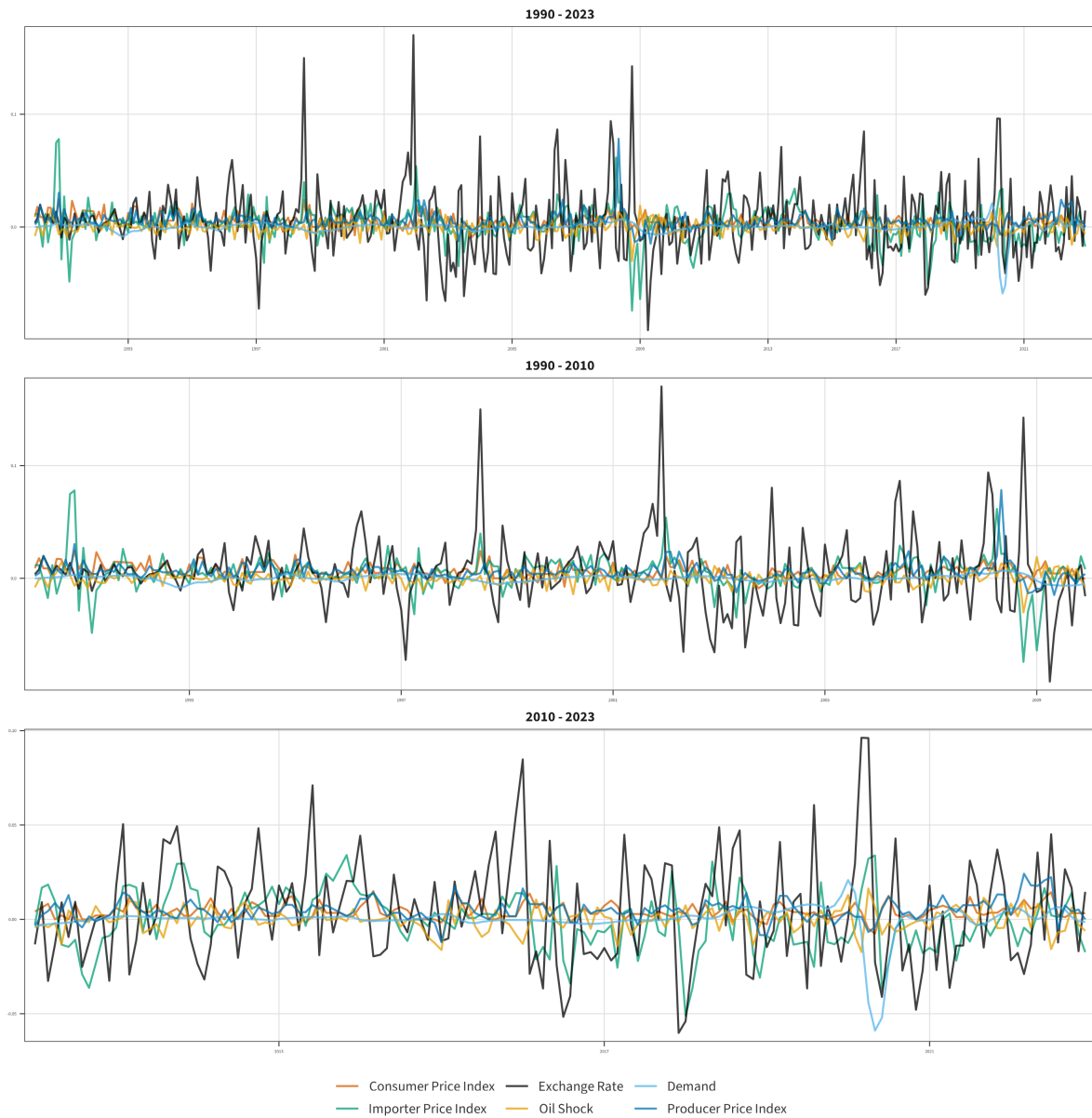


Figure 1: Variable Time Series Plots

Table 4: Augmented Dickey-Fuller Unit Root Tests

Specification	Statistic	1%	5%	10%	Reject
Oil Shock					
Trend	-12.356	-3.98	-3.42	-3.13	Yes
Drift	-12.374	-3.44	-2.87	-2.57	Yes
None	-12.386	-2.58	-1.95	-1.62	Yes
Demand					
Trend	-4.840	-3.98	-3.42	-3.13	Yes
Drift	-4.850	-3.44	-2.87	-2.57	Yes
None	-4.856	-2.58	-1.95	-1.62	Yes
Exchange Rate					
Trend	-13.247	-3.98	-3.42	-3.13	Yes
Drift	-13.243	-3.44	-2.87	-2.57	Yes
None	-12.956	-2.58	-1.95	-1.62	Yes
Import Price Index					
Trend	-8.281	-3.98	-3.42	-3.13	Yes
Drift	-7.859	-3.44	-2.87	-2.57	Yes
None	-7.645	-2.58	-1.95	-1.62	Yes
Producer Price Index					
Trend	-8.352	-3.98	-3.42	-3.13	Yes
Drift	-8.333	-3.44	-2.87	-2.57	Yes
None	-2.746	-2.58	-1.95	-1.62	Yes
Consumer Price Index					
Trend	-4.299	-3.98	-3.42	-3.13	Yes
Drift	-3.953	-3.44	-2.87	-2.57	Yes
None	-1.949	-2.58	-1.95	-1.62	No

Note: ADF test with lag length selected by AIC up to a maximum of 12 lags. H_0 : series contains a unit root. Rejection at the 5% level indicated.

6.2.2 Normality

Table 5: Normality Tests

Variable	Skewness	Kurtosis	JB Stat	p-value
Oil Shock	-0.359	4.632	52.296	p < 0.001
Demand	-6.031	59.492	54919.628	p < 0.001
Exchange Rate	1.047	7.290	374.992	p < 0.001
Import Price Index	-0.130	7.408	320.846	p < 0.001
Producer Price Index	3.061	32.496	14935.865	p < 0.001
Consumer Price Index	0.762	3.792	48.588	p < 0.001

Note: Jarque-Bera test for normality. Skewness and excess kurtosis reported.

6.3 Lag Selection

Table 6: VAR Lag Length Selection Criteria

Lag	AIC	BIC
1	-13412.878	-13233.942
2	-13732.336	-13434.3***
3	-13829.72	-13412.738**
4	-13797.586	-13261.81*
5	-13889.381	-13234.967
6	-13932.879***	-13159.981
7	-13870.065	-12978.839
8	-13904.302*	-12894.904
9	-13928.791**	-12801.377
10	-13893.261	-12647.989
11	-13884.313	-12521.341
12	-13898.35	-12417.837
13	-13843.873	-12245.978
14	-13809.226	-12094.109
15	-13759.976	-11927.796
16	-13702.112	-11753.031
17	-13664.268	-11598.449
18	-13612.239	-11429.842
19	-13542.478	-11243.669
20	-13475.407	-11060.347
21	-13446.789	-10915.644
22	-13375.556	-10728.49
23	-13331.552	-10568.732
24	-13313.852	-10435.443

Note: AIC and BIC computed for VAR with oil shock as exogenous variable. *** minimum, ** second minimum, * third minimum.

6.4 Error Correlation

Table 7: Contemporaneous Correlation Between Errors

	Oil Shock	Demand	Exchange Rate	Import Price Index	Producer Price Index	Consumer Price Index
Oil Shock	1.000	-0.078	0.016	0.017	-0.070	0
Demand	-0.078	1.000	0.141	0.049	0.043	0
Exchange Rate	0.016	0.141	1.000	0.251	0.143	0
Import Price Index	0.017	0.049	0.251	1.000	0.255	0
Producer Price Index	-0.070	0.043	0.143	0.255	1.000	0
Consumer Price Index	0.000	0.000	0.000	0.000	0.000	1

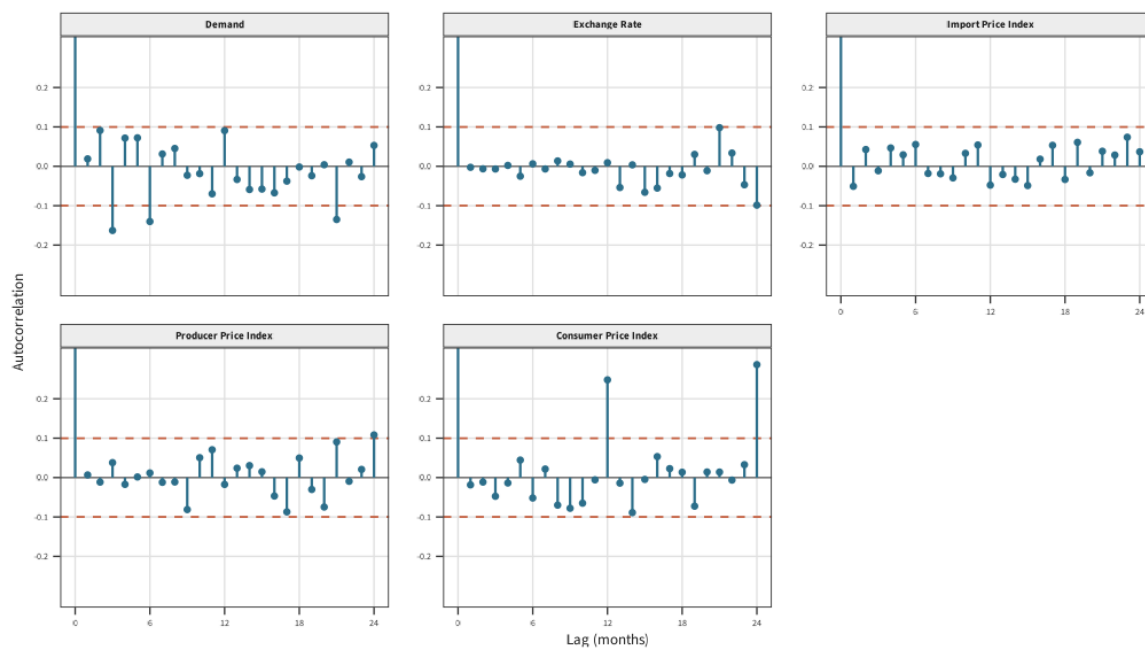


Figure 2: Autocorrelation Functions 1990 - 2023

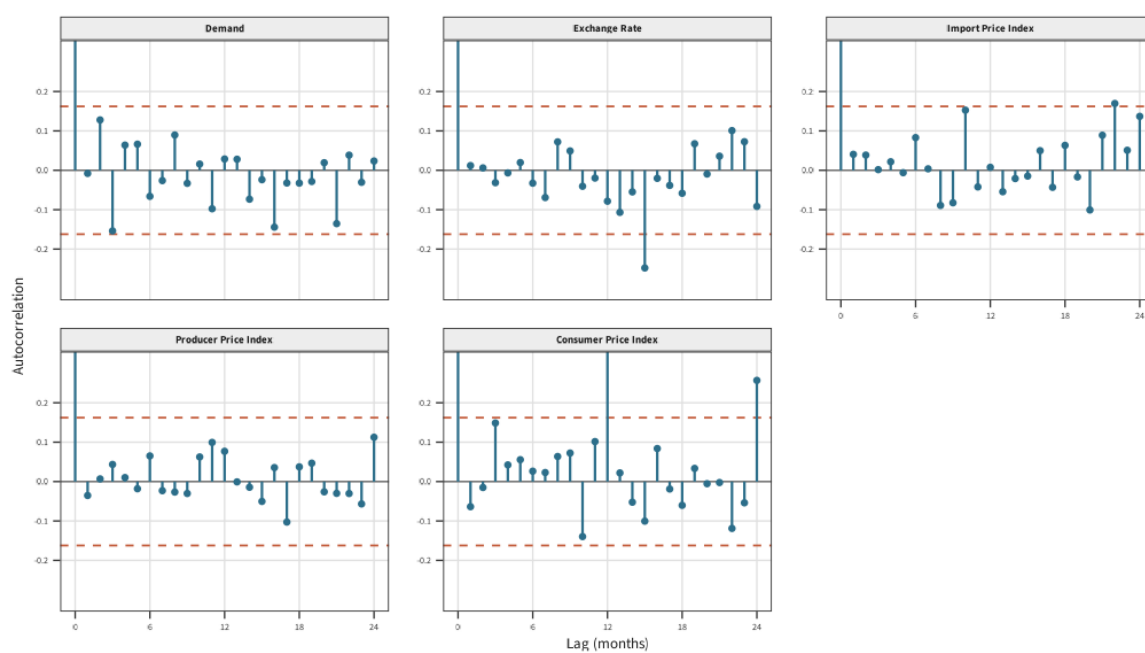


Figure 3: Autocorrelation Functions 2010 - 2023

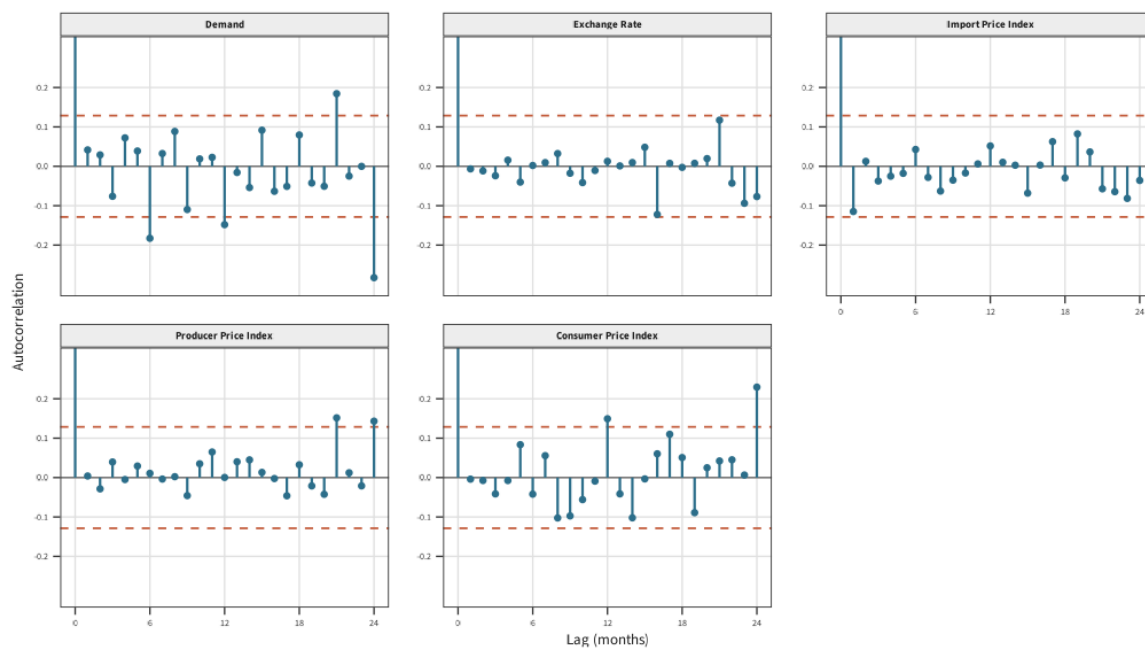


Figure 4: Autocorrelation Functions 1990 - 2010

6.5 Linking Tests

6.5.1 SIC and UVI (Arithmetic)

Table 8: Paired t-test: UVI vs SIC import price series

Mean diff.	t	p	CI low	CI high	df
-0.0028	-0.9088	0.3698	-0.0092	0.0035	34

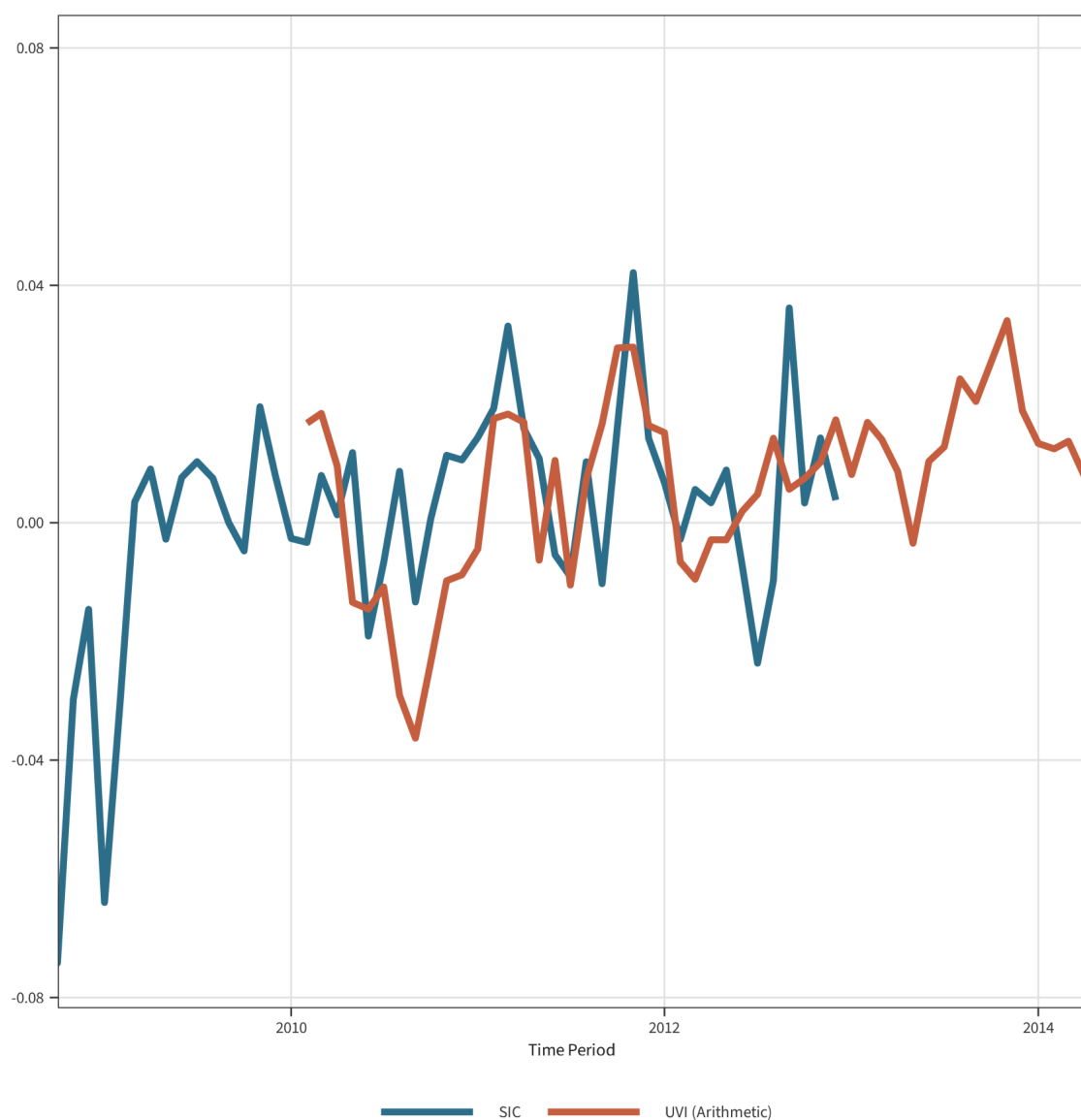


Figure 5: Overlap period for SIC and UVI (Arithmetic) Indices

Table 9: Chow Test for Structural Break at 2010-02

Equation	F-Statistic	p-value
Demand	1.012	0.46
Exchange Rate	0.792	0.856
Import Price Index	2.228	< 0.001***
Producer Price Index	1.204	0.167
Consumer Price Index	1.229	0.142

Note: Chow test for parameter stability at 2010-02-01. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.5.2 UVI (Arithmetic) and UVI (Geometric)

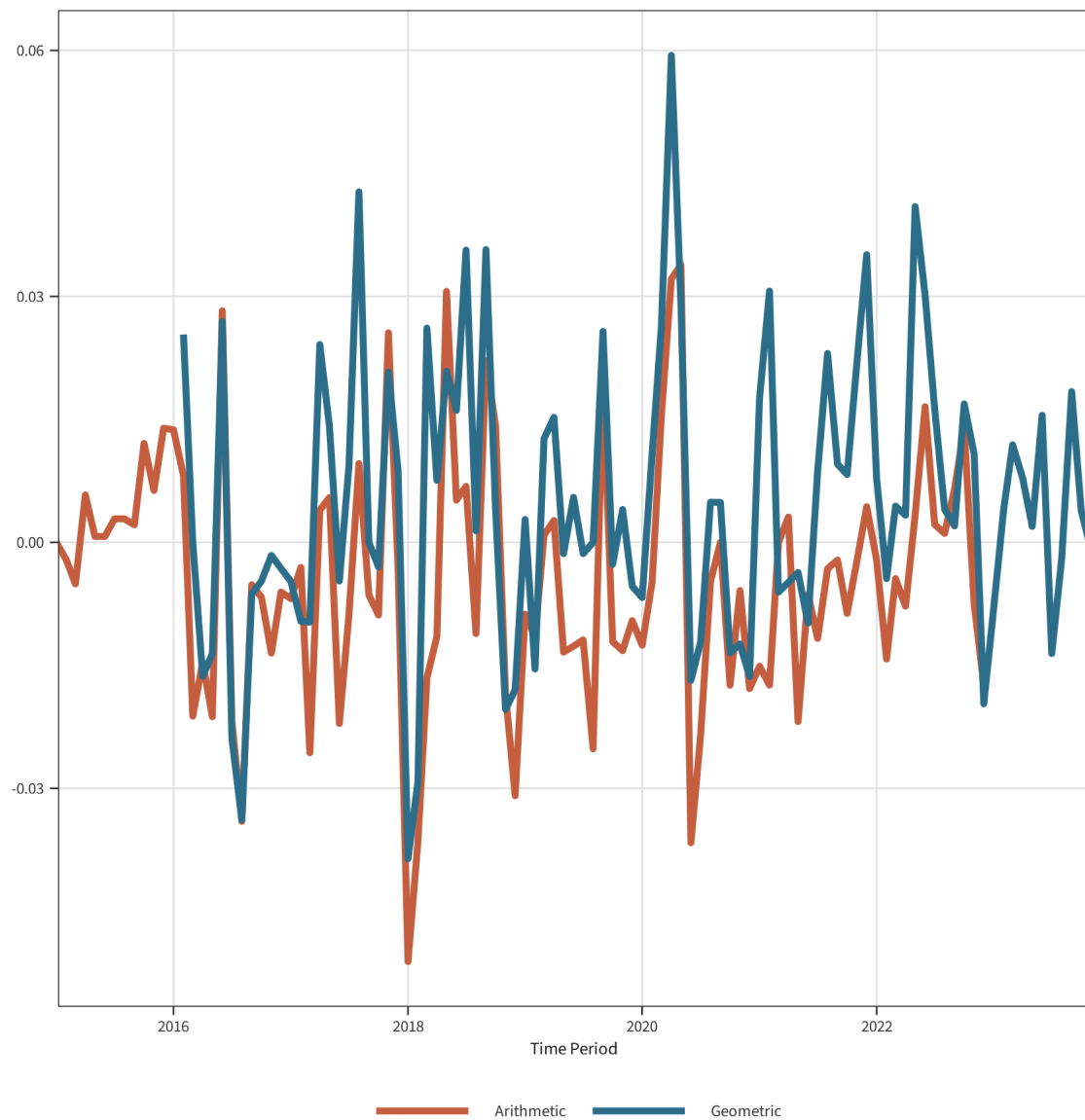


Figure 6: Overlap period for Arithmetic and Geometric UVI Indices

Table 10: Paired t-test: Geometric vs Arithmetic UVI Series

Mean diff.	t	p	CI low	CI high	df
-0.0107	-8.2249	0	-0.0133	-0.0081	82